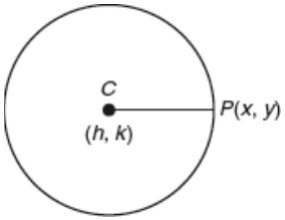
Circle
Part 1

1) Equation of a circle whose centre is (h,k) and radius r

$$(x-h)^2+(y-k)^2=r^2 \qquad \textbf{(1)}$$



Note: If the center of the circle is at the origin, then the equation of the circle is $x^2 + y^2 = r^2$.

2) The general form for equation of a circle

$$x^2+y^2+2mx+2ny+c=0 \qquad \textbf{(2)}$$

This equation can be written in the following form

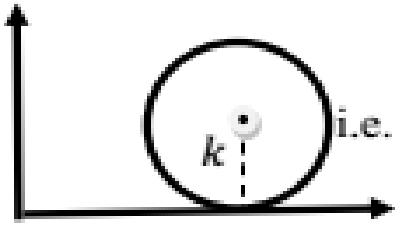
$$(x+m)^2+(y+n)^2=m^2+n^2-c$$

Thus, equation (2) represents a circle whose center is $(-m\ -n)$ and radius $\sqrt{m^2+n^2-c}$

3) Different forms of the equation of the circle:

Equation of a circle with center (h,k) and touches the x-axis

∵ The circle touches the x-axis, that is, r = k. So, the equation of the circle can be in the form



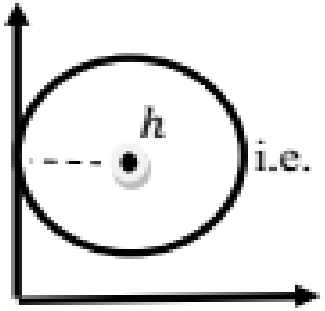
$$(x-h)^2+(y-k)^2=k^2$$

i.e.

$$\boxed{x^2+y^2-2h\,x-2k\,y+h^2=0}$$

Equation of a circle with center (h,k) and touches the y-axis

∵ The circle touches the y-axis, that is, r = h. So, the equation of the circle can be in the form



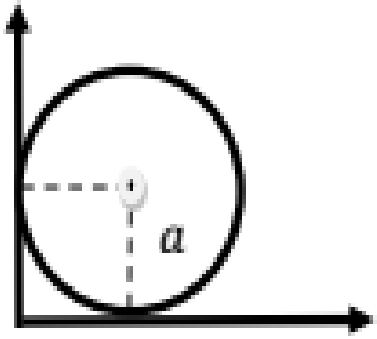
$$(x-h)^2+(y-k)^2=h^2$$

i.e.

$$\boxed{x^2+y^2-2h\,x-2k\,y+k^2=0}$$

● Equation of a circle with radius a and touches the two axes

From the figure we find that the center is (a,a) and the radius is a.



∴ the equation of a circle can be written in the form

$$(x-a)^2+(y-a)^2=a^2$$

i.e.

$$\boxed{x^2+y^2-2a\,x-2a\,y+a^2=0}$$

● Equation of the circle touches the x-axis at the origin

∵ The equation touches the x-axis at the origin

∴ center of the circle is (0,k) and radius is k

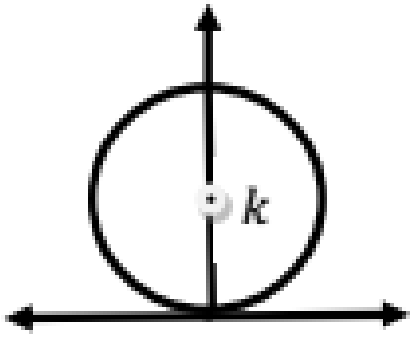
∴ the equation of a circle can be written

in the form

$(x-0)^2+(y-k)^2=k^2$

i.e.

$$\boxed{x^2+y^2-2k\,y=0}$$



● Equation of the circle touches the y-axis at the origin

∵ The equation touches the y-axis at the origin

∴ center of the circle is (h,0) and radius is h

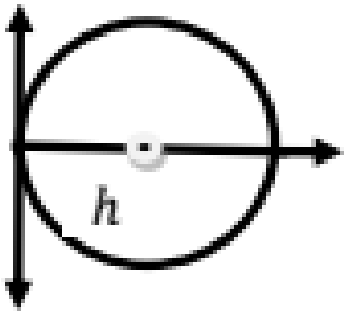
∴ the equation of a circle can be written

in the form

$(x-h)^2+(y-0)^2=h^2$

i.e.

$$\boxed{x^2+y^2-2h\,x=0}$$



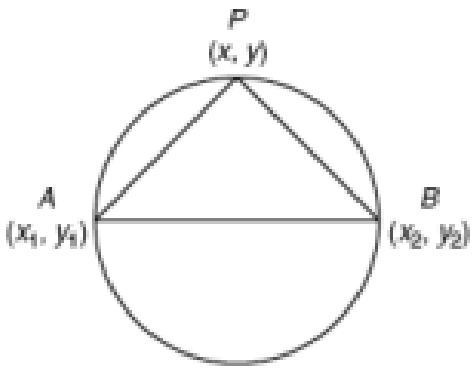
● Equation of circle with the line joining points $A(x_1, y_1)$ and $B(x_2, y_2)$ as the ends of diameter

$A(x_1, y_1)$ and $B(x_2, y_2)$ are the ends of a diameter. Let $P(x, y)$ be any point on the circumference of the circle. Then $AP \perp PB$.

The slope of AP is $m_1 = \frac{y-y_1}{x-x_1}$

The slope of BP is $m_2 = \frac{y-y_2}{x-x_2}$

Since AP is perpendicular to PB, $m_1 m_2 = -1$

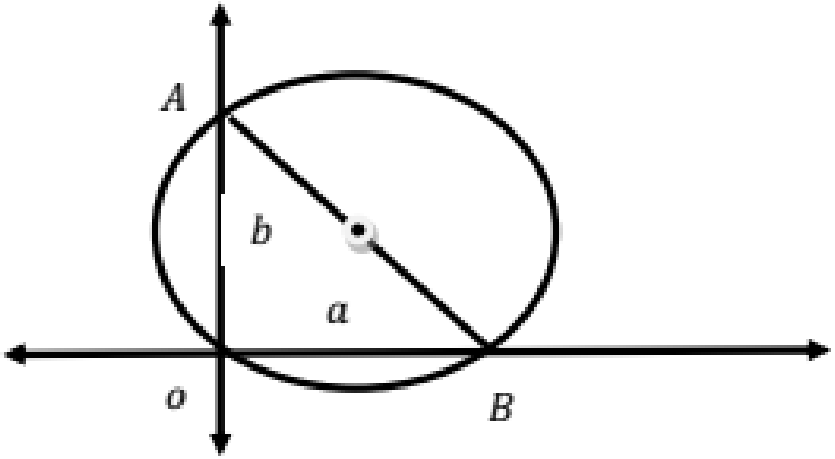


$$\therefore \frac{(y-y_1)}{(x-x_1)} \frac{(y-y_2)}{(x-x_2)} = -1$$

i.e.

$$\boxed{(x-x_1)(x-x_2)+(y-y_1)(y-y_2)=0}$$

● **Equation of a circle that passes through the origin and intersects two coordinates' axes in a and b**



∵ AoB is a right angle and $\overline{AB}$ is a diameter in a circle. ∴ The equation of the circle is

$$(x-a)(x-0)+(y-0)(y-b)=0$$

$$x^2-ax+y^2-by=0$$

Solved Examples:

Example 1

Find the equation of the circle that satisfies the following conditions:

i) Its center is (3,5) and touches the x-axis.

Solution

∵ The circle touches the x-axis so that its radius is 5 and its center is (3,5)

$\therefore\ (x-3)^2+(y-5)^2=(5)^2$

$x^2-6x+9+y^2-10y+25=25$

$$x^2+y^2-6x-10y+9=0$$

iv) It touches the y-axis at the origin and passes through the point (1,2).

Solution

∵ Equation of the circle touches the y-axis at the origin.

$\therefore\ (x-h)^2+(y-0)^2=h^2 \quad x^2+y^2-2hx=0$

$at\ \ (1,2) \quad \Rightarrow \quad 1+4-2h=0 \quad \Rightarrow \quad h=\frac{5}{2}$

$\therefore \quad x^2+y^2-5x=0$

Example 2

Find the center and radius of the following circle

$$x^2 + y^2 + 4x - 10y + 20 = 0$$

Solution

$c = (-m, -n) = (-4/2, 10/2) = (-2, 5)$

$r = \sqrt{m^2 + n^2 - c} = \sqrt{4 + 25 - 20} = \sqrt{9} = 3$

Example 3

Find the equation of the circle in which the two points $(2,-1),(1,3)$ are both ends of its diameter, indicating its center and radius.

Solution

$$(x-x_1)(x-x_2)+(y-y_1)(y-y_2)=0$$

$$(x-2)(x-1)+(y+1)(y-3)=0$$

$$x^2-3x+2+y^2-2y-3=0$$

$$\therefore \quad x^2+y^2-3x-2y-1=0$$

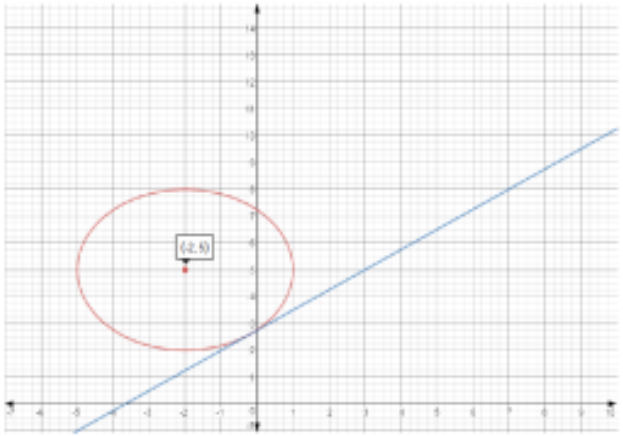
$$c=(-m,-n)=(3/2,1)$$

$$r=\sqrt{m^2+n^2-c}=\sqrt{9/4+1+1}=\sqrt{17/4}=\sqrt{17}/2$$

Find the Equation of the circle for which

vi) Its center is $(-2,5)$ and touches the straight $3x - 4y + 11 = 0$

Solution



The radius is the length of the perpendicular falling from the center of the circle $(-2,5)$ on the straight line $3x - 4y + 11 = 0$

$$r = \left|\frac{ax_1 + b\,y_1 + c}{\sqrt{a^2 + b^2}}\right| = \left|\frac{3(-2) - 4(5) + 11}{\sqrt{9 + 16}}\right|$$

$$= \left|\frac{-6-20+11}{\sqrt{25}}\right| = \frac{15}{5} = 3$$

$$\therefore (x + 2)^2 + (y + 5)^2 = 3^2$$

$$x^2 + 4x + 4 + y^2 + 10y + 25 = 9$$

- **<u>The equation of a circle that passes through three points $(x_1, y_1), (x_2, y_2), (x_3, y_3)$</u>**

Suppose the equation of the circle is

$$x^2 + y^2 + 2mx + 2ny + c = 0$$

Substituting in the equation of the circle with three points, we get three equations in three unknowns, m, n, c, and solving these equations, then substituting for m, n, c in the equation of the circle, we get the equation of the circle.

thank you